

END-TO-END BUSINESS DATA ANALYSIS

Financial Sales & Profit Analysis

Tools Used:

Python (Pandas) | Excel | Power BI

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1. Project Objective

The objective of this project is to analyze financial sales data and identify important business trends related to sales, profit, products, countries, segments, discounts, and yearly performance.

The project follows an end-to-end data analysis process, including:

- Cleaning and preparing the dataset using Python and Pandas.
- Exploring the dataset to identify important patterns and trends.
- Performing data analysis using Pandas and Excel.
- Creating charts and visualizations in Excel.
- Developing an interactive dashboard using Power BI.
- Identifying key business insights from the analysis.
- Providing data-driven recommendations based on the findings.

2. Tools & Technologies

The following tools and technologies were used to complete the project:

Tool / Technology	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation and exploratory data analysis
Excel	Data analysis, PivotTables, and charts
Power BI	Interactive dashboard and data visualization
CSV	Storage and transfer of the cleaned dataset

Dataset

The dataset contains **700 rows and 16 columns** covering financial sales information.

The main fields include:

- Segment
- Country
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price
- Gross Sales
- Discounts
- Sales
- COGS
- Profit
- Date
- Month Number
- Month Number
- Year

3. Data Cleaning Using Python

The dataset was cleaned and prepared using Python and the Pandas library before performing the analysis.

Data Cleaning Steps

1. **Loaded the dataset** into a Pandas DataFrame.
2. **Checked the dataset structure** using rows, columns, and data types.
3. **Cleaned column names** by removing unnecessary spaces.
4. **Checked for missing values** in all columns.
5. The missing values in the **Discount Band** column were replaced with "**No Discount**".
6. **Checked for duplicate records** and found no duplicate rows.
7. Converted the **Date** column into the appropriate datetime format.
8. Verified the final data types and dataset structure.
9. Saved the cleaned dataset as **cleaned_data.csv**.

Final Dataset

After cleaning, the dataset contained:

- **700 rows**
- **16 columns**
- **0 missing values**
- **0 duplicate rows**

The cleaned dataset was then used for exploratory analysis, Excel visualization, and the Power BI dashboard.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python and Pandas to understand the financial dataset and identify important business patterns.

Key Analysis Performed

The following areas were analyzed:

- Total Sales and Total Profit
- Average Sales and Average Profit
- Highest and lowest profit records
- Sales and Profit by Product
- Sales and Profit by Country
- Sales and Profit by Segment
- Sales and Profit by Discount Band
- Sales and Profit by Year
- Units Sold by Product and Country

Important Findings

- **Paseo** generated the highest sales and profit among the products.
- **Carretera** had the lowest sales and profit among the products.
- **2014** recorded substantially higher sales and profit than 2013.
- The **Medium Discount Band** recorded the highest sales.
- The **Low Discount Band** recorded the highest profit.
- The **High Discount Band** had the highest total discount amount.
- The **No Discount** category had the lowest sales.

These findings were further visualized using Excel and Power BI.

5. Excel Analysis & Visualizations

Analysis Performed

- Product Analysis
- Country Analysis
- Segment Analysis
- Discount Band Analysis
- Year Analysis

Charts Created

- Sales by Product
- Profit by Product
- Sales by Country
- Sales by Segment
- Sales & Profit by Year
- Sales & Profit by Discount Band

Key Insight

Paseo generated the highest sales and profit among the products, while **Carretera** recorded the lowest sales and profit.

6. Power BI Dashboard

An interactive dashboard was created using Microsoft Power BI to visualize the key business metrics and provide an easy-to-understand overview of the financial data.

Key Performance Indicators

The dashboard contains four KPI cards:

- **Total Sales**
- **Total Profit**
- **Total Units Sold**
- **Total Discounts**

Visualizations

The dashboard includes:

- Sales by Product
- Profit by Product
- Sales by Country
- Sales by Segment
- Sales & Profit by Year
- Sales & Profit by Discount Band

Interactive Filter

A **Year slicer** was added to allow users to filter the dashboard by year. Selecting a year updates the dashboard visuals accordingly.

The Power BI dashboard provides an interactive way to explore business performance and compare different products, countries, segments, discount bands, and years

7. Key Findings

The analysis of the financial dataset produced the following key findings:

1. **Product Performance:** Paseo generated the highest sales and profit among the products, while Carretera recorded the lowest sales and profit.
2. **Yearly Performance:** 2014 recorded significantly higher sales and profit compared with 2013.
3. **Discount Analysis:** The Medium Discount Band recorded the highest sales, while the Low Discount Band recorded the highest profit.
4. **Discount Amount:** The High Discount Band had the highest total discount amount.
5. **Sales Distribution:** Sales varied across countries and business segments, indicating differences in market performance.
6. **Profitability:** Products and discount categories showed different levels of profitability, highlighting the importance of analyzing both sales and profit rather than sales alone.

The analysis provides a clear view of sales and profitability patterns across products, countries, segments, discount bands, and years.

8. Business Recommendations

Based on the analysis, the following recommendations can be considered:

1. **Focus on High-Performing Products:** Products such as Paseo that generate strong sales and profit can receive greater attention in sales and marketing activities.
2. **Review Low-Performing Products:** Products with comparatively lower sales and profit should be reviewed to understand customer demand, pricing, and market conditions.
3. **Monitor Discount Strategies:** Different discount bands show different sales and profit patterns. Discount levels should therefore be monitored carefully to maintain profitability.
4. **Analyze High-Discount Areas:** The High Discount Band recorded the highest discount amount. The impact of these discounts on profitability should be monitored before increasing discount levels.
5. **Study Yearly Trends:** The strong performance in 2014 can be examined further to identify the factors contributing to higher sales and profit.
6. **Use Interactive Dashboards:** The Power BI dashboard can be used by stakeholders to monitor performance and explore trends by year, product, country, segment, and discount band.

9. Conclusion

This project demonstrated an end-to-end approach to business data analysis using Python, Pandas, Excel, and Power BI.

The dataset was successfully cleaned and explored using Python. Pandas was used to identify important patterns in sales, profit, products, countries, segments, discount bands, and yearly performance.

Excel was used to create PivotTables and charts for detailed analysis, while Power BI was used to develop an interactive dashboard containing key performance indicators, charts, and a year filter.

The analysis provided useful insights into business performance and demonstrated how data visualization can help present complex financial information in a clear and understandable manner.

This project strengthened practical skills in **data cleaning, exploratory data analysis, Excel, data visualization, Power BI, and business insight generation.**